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Question

Consider a discrete-time LTI system with input $x[n] = \delta[n] + \delta[n-1]$ and impulse response $h[n] = \delta[n] - \delta[n-1]$.
The output of the system will be given by:

(a) $\delta[n] - \delta[n-2]$

(b) $\delta[n] - \delta[n-1]$

(c) $\delta[n-1] + \delta[n-2]$

(d) $\delta[n] + \delta[n-1] + \delta[n-2]$

Solution

For an LTI system, the output is obtained by convolution:

$$y[n] = x[n] * h[n]. \tag{1}$$

Given input and impulse response:

$$x[n] = \delta[n] + \delta[n-1], \tag{2}$$
$$h[n] = \delta[n] - \delta[n-1]. \tag{3}$$

Representing them as vectors:

$$\mathbf{x} = \begin{pmatrix} x[0] \\ x[1] \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{h} = \begin{pmatrix} h[0] \\ h[1] \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{4}$$

Determine output length:

$$N_y = N_x + N_h - 1 = 2 + 2 - 1 = 3. \tag{5}$$

Solution

Form the convolution (Toeplitz) matrix using $\mathbf{h}$:

$$\mathbf{H} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix}. \tag{6}$$

Compute the output vector:

$$\begin{aligned} \mathbf{y} &= \mathbf{Hx} \\ &= \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1(1) + 0(1) \\ -1(1) + 1(1) \\ 0(1) - 1(1) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}. \end{aligned} \tag{7}$$

Thus, the output samples are:

$$y[0] = 1, \qquad y[1] = 0, \qquad y[2] = -1. \tag{8}$$

Hence, the output signal is:

$$y[n] = \delta[n] - \delta[n-2]. \tag{9}$$

Final answer:

$$\boxed{y[n] = \delta[n] - \delta[n-2] \quad \text{(Option (a))}} \tag{10}$$